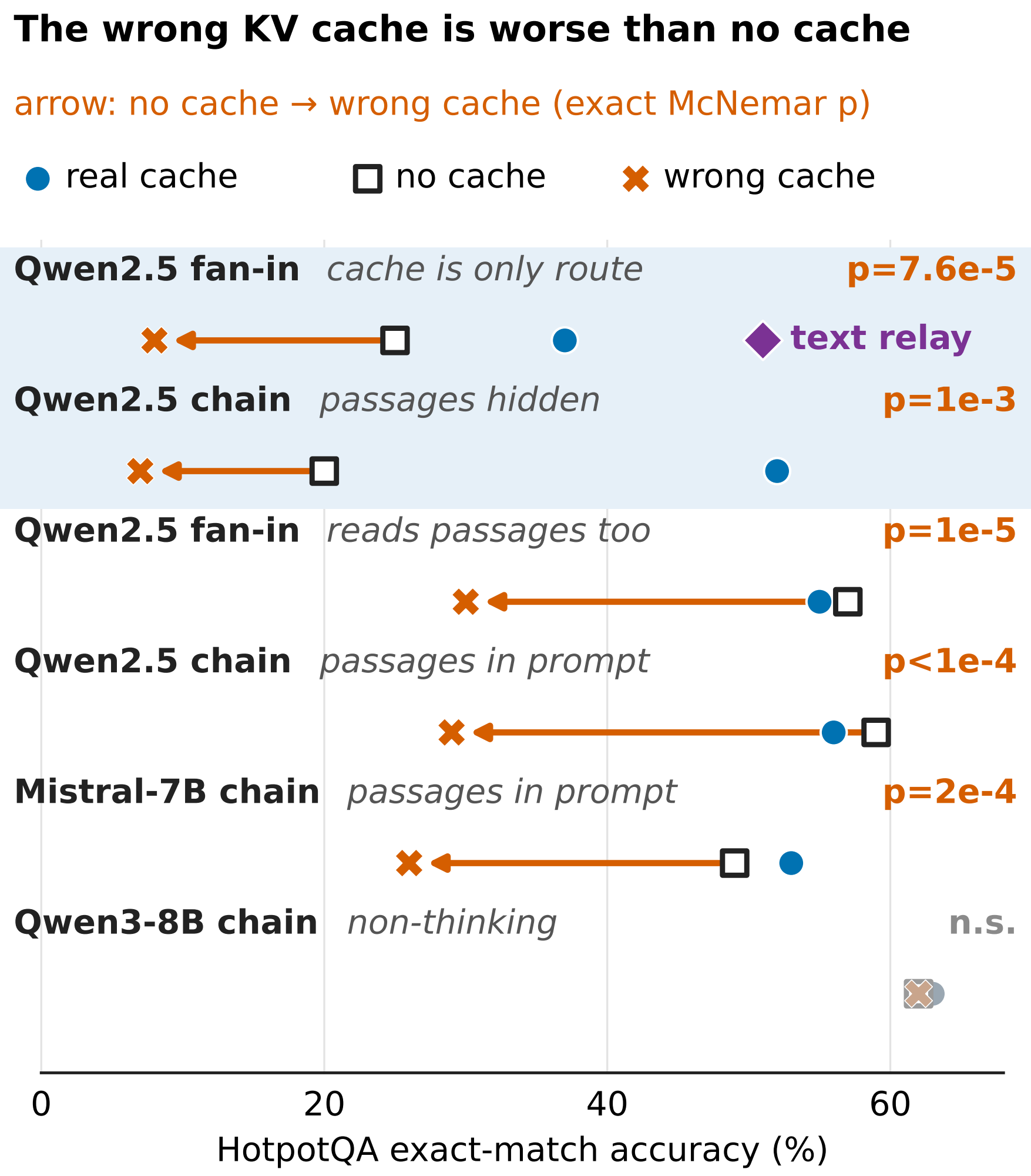
The wrong KV cache is worse than no cache
arrow: no cache → wrong cache (exact McNemar p)
real cache
no cache
wrong cache
Qwen2.5 fan-in cache is only route
p=7.6e-5
text relay
Qwen2.5 chain passages hidden
p=1e-3
Qwen2.5 fan-in reads passages too
p=1e-5
Qwen2.5 chain passages in prompt
p<1e-4
Mistral-7B chain passages in prompt
p=2e-4
Qwen3-8B chain non-thinking
n.s.
0
20
40
60
HotpotQA exact-match accuracy (%)